

Q1 - 25 July - Shift 2

A drop of liquid of density ρ is floating half immersed in a liquid of density σ and surface tension $7.5 \times 10^{-4} \text{ Ncm}^{-1}$. The radius of drop in cm will be : (Take : $g = 10 \text{ m/s}^2$)

(A) $\frac{15}{\sqrt{2\rho - \sigma}}$

(B) $\frac{15}{\sqrt{\rho - \sigma}}$

(C) $\frac{3}{2\sqrt{\rho - \sigma}}$

(D) $\frac{3}{20\sqrt{2\rho - \sigma}}$

Space for your notes:

Q2 - 26 July - Shift 1

A water drop of radius 1 cm is broken into 729 equal droplets. If surface tension of water is 75 dyne/cm, then the gain in surface energy upto first decimal place will be :

[Given $\pi = 3.14$]

(A) $8.5 \times 10^{-4} \text{ J}$

(B) $8.2 \times 10^{-4} \text{ J}$

(C) $7.5 \times 10^{-4} \text{ J}$

(D) $5.3 \times 10^{-4} \text{ J}$

Official Ans. by NTA (C)

Q3 - 27 July - Shift 1

Questions

MathonGo

Two cylindrical vessels of equal cross-sectional area 16 cm^2 contain water upto heights 100 cm and 150 cm respectively. The vessels are interconnected so that the water levels in them become equal. The work done by the force of gravity during the process, is [Take density of water $= 10^3 \text{ kg/m}^3$ and $g = 10 \text{ ms}^{-2}$]

- (A) 0.25 J (B) 1 J
(C) 8 J (D) 12 J

*Space for your notes:***Q4 - 27 July - Shift 2**

A spherical soap bubble of radius 3 cm is formed inside another spherical soap bubble of radius 6 cm . If the internal pressure of the smaller bubble of radius 3 cm in the above system is equal to the internal pressure of the another single soap bubble of radius $r \text{ cm}$. The value of r is.....

*Space for your notes:***Q5 - 28 July - Shift 1**

The diameter of an air bubble which was initially 2 mm , rises steadily through a solution of density 1750 kg m^{-3} at the rate of 0.35 cms^{-1} . The coefficient of viscosity of the solution is _____ poise (in nearest integer). (the density of air is negligible).

Space for your notes:

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Q6 - 28 July - Shift 2

A pressure-pump has a horizontal tube of cross-sectional area 10 cm^2 for the outflow of water at a speed of 20 m/s . The force exerted on the vertical wall just in front of the tube which stops water horizontally flowing out of the tube, is:

[given : density of water = 1000 kg/m^3]

(A) 300 N

(B) 500 N

(C) 250 N

(D) 400 N

Space for your notes:

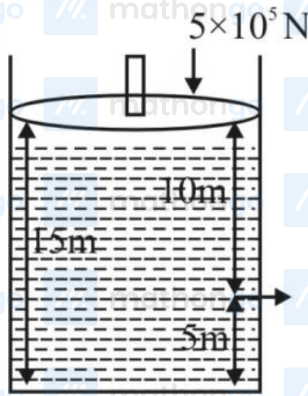
Q7 - 28 July - Shift 2

Questions

MathonGo

Consider a cylindrical tank of radius 1m is filled with water. The top surface of water is at 15 m from the bottom of the cylinder. There is a hole on the wall of cylinder at a height of 5m from the bottom. A force of 5×10^5 N is applied on the top surface of water using a piston. The speed of efflux from the hole will be :

(given atmospheric pressure $P_A = 1.01 \times 10^5$ Pa, density of water $\rho_w = 1000 \text{ kg/m}^3$ and gravitational acceleration $g = 10 \text{ m/s}^2$)



- (A) 11.6 m/s (B) 10.8 m/s
(C) 17.8 m/s (D) 14.4 m/s

Q8 - 29 July - Shift 1

Questions

MathonGo

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Clothes containing oil or grease stains cannot be cleaned by water wash.

Reason (R) : Because the angle of contact between the oil/ grease and water is obtuse. In the light of the above statements, choose the correct answer from the option given below.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is true but (R) is true

Space for your notes:

Q9 - 29 July - Shift 2

The velocity of a small ball of mass 0.3g and density 8g/cc when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is 1.3 g/cc, then the value of viscous force acting on the ball will be

$x \times 10^{-4}$ N, the value of x is _____. [use $g = 10\text{m/s}^2$]

Space for your notes:

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Answer Key

Q1 (A)

Q2 (C)

Q3 (B)

Q4 (2)

Q5 (11)

Q6 (D)

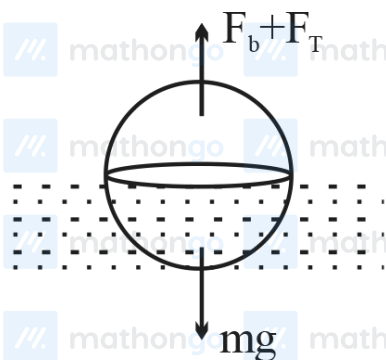
Q7 (C)

Q8 (A)

Q9 (25)

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Q1 (A)



Boyant force + surace tension = mg

$$\sigma \frac{V}{2} g + 2\pi R T = \rho V g$$

$$2\pi R T = \frac{(2\rho - \sigma)}{2} \cdot \frac{4}{3} \pi R^3 g; \left[V = \frac{4}{3} \pi R^3 \right]$$

$$R^3 = \frac{3T}{(2\rho - \sigma)g} \Rightarrow R = \sqrt{\frac{3 \times 7.5 \times 10^{-2} \text{ N-m}^{-1}}{(2\rho - \sigma) \times 10}}$$

$$R = \frac{3}{20\sqrt{(2\rho - \sigma)}} \text{ m} = \frac{15}{\sqrt{2\rho - \sigma}} \text{ cm}$$

Q2 (C)

Initial surface energy = TA

Where T is surface tension and A is surface area

$$U_i = \left(\frac{75 \times 10^{-5} \text{ N}}{10^{-2} \text{ m}} \right) \times \left[4\pi (1 \times 10^{-2})^2 \right]$$

$$= 75 \times 10^{-3} \times 4\pi \times 10^{-4} = 942 \times 10^{-7} \text{ J}$$

To get final radius of drops by volume

conservation

$$\frac{4}{3}\pi R^3 = 729 \left(\frac{4}{3}\pi r^3 \right)$$

R = Initial radius

r = final radius

$$r = \frac{R}{(729)^{1/3}} = \frac{R}{9} = \frac{1}{9} \text{ cm}$$

Final surface energy

$$U_f = 729 [TA]$$

$$= 729 \left[\frac{75 \times 10^{-5} \text{ N}}{10^{-2} \text{ m}} \right] \times \left[4\pi \left(\frac{1}{9} \times 10^{-2} \right)^2 \right]$$

$$= 729 \left[75 \times 10^{-3} \times \frac{4\pi \times 10^{-4}}{81} \right]$$

$$= 9 [942 \times 10^{-7} \text{ J}]$$

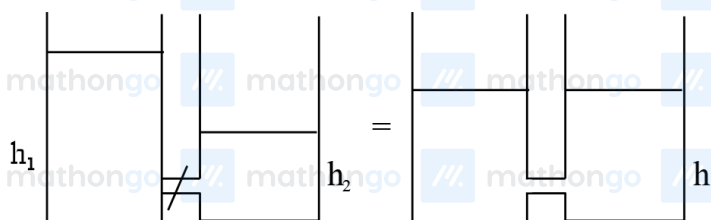
Gain in surface energy

$$\Delta U = 9 \times 942 \times 10^{-7} - 942 \times 10^{-7}$$

$$= 8 \times 942 \times 10^{-7} \text{ J} = 7536 \times 10^{-7} \text{ J}$$

Q3 (B)

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$$h = \frac{h_1 + h_2}{2}$$

Now,

$$W = U_i - U_f$$

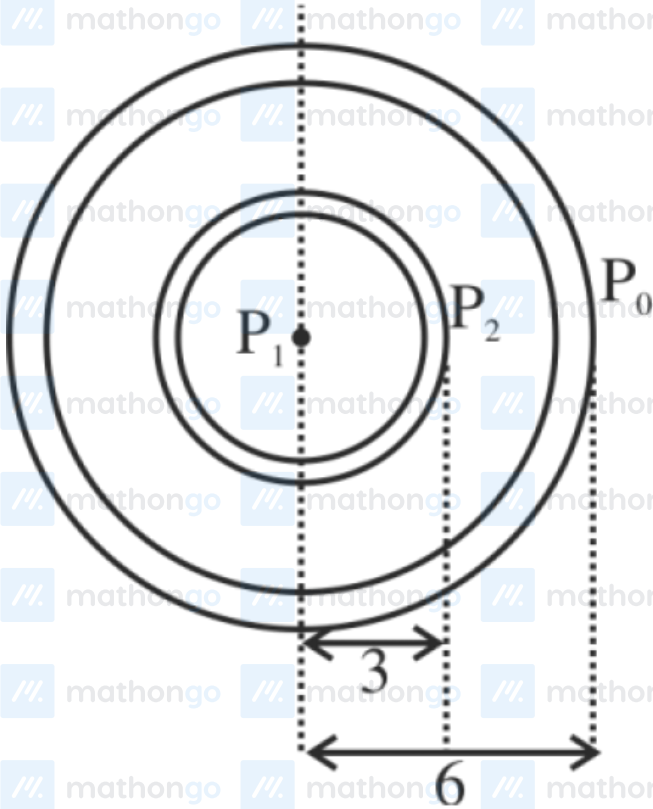
$$W = (\rho A h_1) g \frac{h_1}{2} + (\rho A h_2) g \frac{h_2}{2} - \rho A (h_1 + h_2) g$$

$$\left(\frac{h_1 + h_2}{4} \right)$$

$$W = \frac{\rho A g}{2} \left[h_1^2 + h_2^2 - \frac{(h_1 + h_2)^2}{2} \right]$$

$$W = 1 \text{ J}$$

Q4 (2)

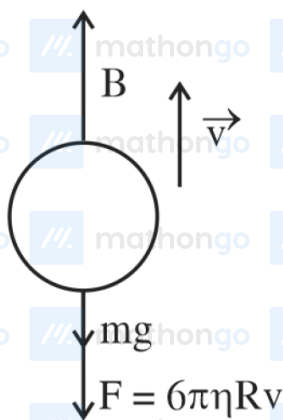


$$P_2 - P_0 = \frac{4T}{6} \text{ \& } P_1 - P_2 = \frac{4T}{3}$$

$$\Rightarrow P_1 - P_0 = \frac{4T}{2} = 2$$

Q5 (11)

As the bubble is rising steadily the net force acting on it will be zero



(Because of density of air the value of mg can be neglected)

$$\text{So } B = F \Rightarrow \frac{4\pi}{3} R^3 \rho g = 6\pi\eta R v$$

Putting $R = 1 \text{ mm} = 10^{-3} \text{ m}$

$$\rho = 1.75 \times 10^3 \text{ kg/m}^3$$

$$g = 10 \text{ m/s}^2$$

$$v = 0.35 \times 10^{-2} \text{ m/s}$$

$$\eta = \frac{10}{9} \approx 1.11 \text{ SI unit} = 11 \text{ poise (CGS)}$$

Q6 (D)

$$F = \rho a v^2 = 10^3 \times 10 \times 10^{-4} \times 20 \times 20$$

$$F = 400$$

Q7 (C)

Apply Bernoulli's theorem between Piston and

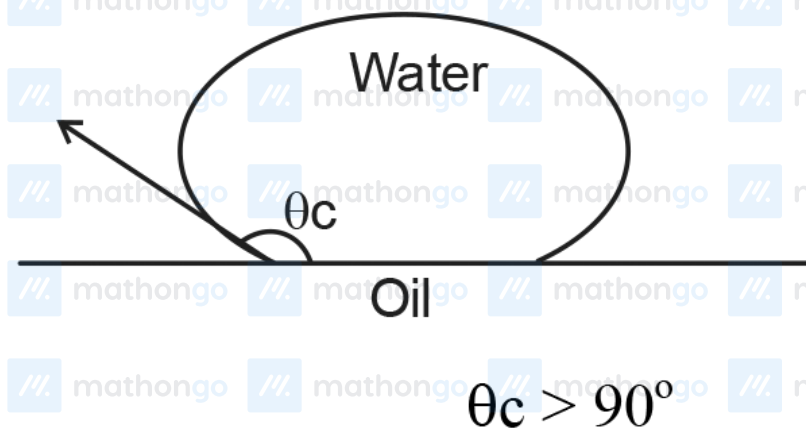
hole $P_A + \rho gh = P_0 + \frac{1}{2} \rho v_e^2$

Assuming there is no atmospheric pressure on piston

$$\frac{5 \times 10^5}{\pi} + 10^3 \times 10 \times 10 = 1.01 \times 10^5 + \frac{1}{2} \times 10^3 \times v_e^2$$

$$v_e = 17.8 \text{ m/s}$$

Q8 (A)



For water oil interface

Q9 (25)

$$F_V + F_B = mg \quad (v = \text{constant})$$

$$F_V = mg - F_B$$

$$= \rho_B Vg - \rho_L Vg$$

$$= (\rho_B - \rho_L) Vg$$

$$= (8 - 1.3) \times 10^3 \times \frac{0.3 \times 10^{-3}}{8 \times 10^3} \times 10$$

$$= \frac{6.7 \times 0.3}{8} \times 10^{-2} \quad (g = 10)$$

$$= \frac{67 \times 3}{8} \times 10^{-4} = 25.125 \times 10^{-4}$$

Ans. 25.125